

Employee Attrition Prediction using Machine Learning

Executive Summary

Project Overview

Employee attrition can significantly impact an organization's productivity, recruitment costs, and overall business performance. The objective of this project was to analyse employee data and identify the key factors that influence whether an employee is likely to leave the company. By understanding these factors, the Human Resources (HR) department can take proactive steps to improve employee retention and reduce turnover.

Key Findings

Analysis of the IBM HR Analytics dataset revealed that approximately **16%** of employees left the organization, indicating that while most employees remain with the company, a significant number are at risk of leaving. Employees working in the **Sales department** experienced the highest attrition rate, with **Sales Representatives** among the job roles showing the greatest employee turnover.

The analysis also showed that employees with **lower monthly income**, **frequent overtime**, **shorter tenure**, and **poorer work-life balance** were more likely to leave the organization. These factors collectively provide valuable insights into employee retention and workforce planning.

Predictive Analysis

Three machine learning models were developed and compared to predict employee attrition:

- Logistic Regression
- Random Forest Classifier
- Gradient Boosting Classifier

Among these models, the **Random Forest Classifier** provided the most accurate and reliable predictions for identifying employees who are at higher risk of leaving the company.

Recommendations for HR

Based on the findings, the following actions are recommended:

- Focus retention programs on employees in the Sales department and other high-risk job roles.
- Improve work-life balance by reducing excessive overtime and monitoring employee workload.
- Provide career development opportunities and regular feedback sessions, especially for employees during their first few years with the company.
- Review compensation and recognition programs for employees with lower salaries to improve job satisfaction and retention.

Project Limitation

The prediction model is based on historical employee data and identifies patterns rather than definite reasons for employee resignation. Personal circumstances, external job opportunities, and organizational changes are not included in the dataset. Therefore, the model should be used as a decision-support tool alongside HR expertise rather than as the sole basis for making employee-related decisions.

Conclusion

This project demonstrates how data-driven analysis can help organizations better understand employee behaviour and improve retention strategies. By identifying employees who may be at risk of leaving, HR teams can take timely and targeted actions to enhance employee satisfaction, reduce turnover, and support long-term organizational growth.